

An Autonomous 3-Point Harmonic Discrete Laplacian Lattice:

Part I: Tri-Axial Geometric Rigidity, Apéry Condensation, and Operator Kernel Triviality

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Abstract

We formulate the rigorous mathematical foundations of an infinite-dimensional discrete kinetic lattice defined on the sequence Hilbert space $\ell^2(\mathbb{N}_{\geq 2})$. Governed by the discrete second central difference of the harmonic sequence, $w(k) \equiv 2\Delta^2(1/k) = \frac{4}{k(k^2-1)}$, the framework establishes three exact, non-perturbative geometric invariants: (i) an exact telescoping partition of unity ($\sum_{k=2}^{\infty} w(k) \equiv 1$) displaying exact 4-fold homomorphic scaling with the odd-zeta residue measure $\sum_{m=1}^{\infty} (\zeta(2m+1) - 1) = 1/4$; (ii) a transverse radial curvature operator $\Delta_{\perp}^2(K) = \frac{(K-1)^2}{K^{K+1}}$ anchoring the base geometry to the exact rational curvature $\Delta_{\perp}^2(2) = 1/8 = 0.125$ while constructing an impenetrable super-exponential ultraviolet dissipation wall ($e^{-K \ln K}$); and (iii) an Apéry vacuum trace condensation wherein the diagonal self-energy operator $\mathcal{Z}_{\text{diag}} \equiv \text{Tr}(\hat{W}_w)$ exhibits complete algebraic anti-symmetric cancellation of even-order Basel terms ($\zeta(2) = \pi^2/6$), condensing into the closed form $4\zeta(3) - 19/4 \approx 0.058228$ and pinning 99.39% of the state measure into the fundamental doublet $\{2, 3\}$. By applying the Phragmén-Lindelöf principle and Carlson's theorem under a half-plane shift, we prove the strict triviality of the transfer operator point spectrum ($\ker \hat{T}_w(s) = \{0\}$), establishing the absence of non-trivial bound states across the critical strip. Finally, arithmetic frustration gaps bounded via Baker's theorem guarantee spectral non-degeneracy. This establishes the discrete structural foundation for the dual-track continuous boundary spectral audit presented in Part II.

Keywords: Harmonic Discrete Laplacian, Telescoping Unit Measure, Apéry Trace Condensation, Carlson Kernel Triviality, Transverse Curvature Anchor, Boundary-Coupled Spectral Synthesis.

1 Introduction

A central open problem in spectral geometry and quantum chaos is formulating an autonomous, self-adjoint microscopic lattice that naturally reflects the arithmetic distribution of continuous spectra [1]. While heuristic Hamiltonian models often encounter spectral divergence or rely on ad-hoc regularization, discrete sequence spaces provide an intrinsic algebraic framework free of ultraviolet infinities [2].

In this paper, Part I of a three-part research suite, we construct the rigorous operator-theoretic foundation of the **3-Point Harmonic Discrete Laplacian Lattice** on $\ell^2(\mathbb{N}_{\geq 2})$. Rather than imposing continuous boundary conditions a priori, we derive the kinetic coupling weights directly from the discrete second central difference of the harmonic potential $1/k$.

The resulting framework is anchored by three foundational algebraic pillars:

1. **Metric Rigidity and Unit Measure (Axis I):** The discrete weight $w(k) = \frac{4}{k(k^2-1)}$ telescopes identically to unity ($\sum_{k=2}^{\infty} w(k) \equiv 1$), mirroring the odd-zeta residue measure.
2. **Transverse Curvature Anchor (Axis II):** A radial shell curvature metric $\Delta_{\perp}^2(K) = \frac{(K-1)^2}{K^{K+1}}$ establishes an exact rational anchor $\Delta_{\perp}^2(2) = 1/8 = 0.125$ and erects a super-exponential UV barrier.
3. **Apéry Trace Condensation (Axis III):** The diagonal self-energy trace $\mathcal{Z}_{\text{diag}} \equiv \text{Tr}(\hat{W}_w)$ cancels all $\zeta(2)$ contributions identically, condensing into the closed analytic form $4\zeta(3) - 19/4$ and pinning 99.39% of the bulk measure into the fundamental doublet $\{2, 3\}$.

Furthermore, applying Carlson's theorem to the weighted transfer operator $\hat{T}_w(s)$ proves strict kernel triviality ($\ker \hat{T}_w(s) = \{0\}$), ruling out spurious bound states across the critical strip. These theorems provide the bedrock for the boundary-coupled spectral synthesis and 105-DPS numerical audit carried out in Part II.

2 Axis I: Discrete Metric Rigidity and Partition of Unity

Let $\mathcal{H}_0 = \ell^2(\mathbb{N}_{\geq 2})$ be the separable Hilbert space of square-summable sequences over natural indices $k \geq 2$, endowed with the standard inner product $\langle u, v \rangle = \sum_{k=2}^{\infty} \overline{u_k} v_k$.

Definition 2.1 (Harmonic Discrete Laplacian Weight). The kinetic coupling weight $w(k)$ is defined as twice the second central difference of the discrete harmonic sequence $h(k) = 1/k$:

$$w(k) \equiv 2\Delta^2\left(\frac{1}{k}\right) = 2\left[\frac{1}{k-1} - \frac{2}{k} + \frac{1}{k+1}\right] = \frac{4}{k(k^2-1)}. \quad (1)$$

Theorem 2.2 (Telescoping Partition of Unity). *The sequence $\{w(k)\}_{k=2}^{\infty}$ forms a strict probability measure on $\mathbb{N}_{\geq 2}$:*

$$\sum_{k=2}^{\infty} w(k) \equiv 1. \quad (2)$$

Under finite modal truncation at cutoff N , the residual boundary leakage contracts as:

$$\mathcal{E}_N \equiv 1 - \sum_{k=2}^N w(k) = \frac{2}{N(N+1)} = \mathcal{O}(N^{-2}). \quad (3)$$

Proof. Rewriting the central difference in partial fractions:

$$w(k) = 2\left[\left(\frac{1}{k-1} - \frac{1}{k}\right) - \left(\frac{1}{k} - \frac{1}{k+1}\right)\right] = 2\left[\frac{1}{(k-1)k} - \frac{1}{k(k+1)}\right]. \quad (4)$$

Letting $A_k = \frac{2}{k(k-1)}$, the sum telescopes directly:

$$\sum_{k=2}^N w(k) = \sum_{k=2}^N (A_k - A_{k+1}) = A_2 - A_{N+1} = 1 - \frac{2}{N(N+1)}. \quad (5)$$

Taking $N \rightarrow \infty$, the boundary remainder $\mathcal{E}_N \rightarrow 0$ monotonically, proving exact unit measure closure. $\square$

Theorem 2.3 (Odd-Zeta Residue Homomorphism). *The total lattice measure is strictly 4-fold homomorphic to the infinite sum of the odd-zeta residues:*

$$\sum_{k=2}^{\infty} w(k) = 4 \sum_{m=1}^{\infty} (\zeta(2m+1) - 1) = 4 \times \frac{1}{4} \equiv 1. \quad (6)$$

Proof. Expanding the geometric series for $w(k)$:

$$w(k) = \frac{4}{k^3(1 - 1/k^2)} = 4 \sum_{m=1}^{\infty} \frac{1}{k^{2m+1}}. \quad (7)$$

Summing over all bulk lattice modes $k \geq 2$:

$$\sum_{k=2}^{\infty} w(k) = 4 \sum_{m=1}^{\infty} \sum_{k=2}^{\infty} \frac{1}{k^{2m+1}} = 4 \sum_{m=1}^{\infty} (\zeta(2m+1) - 1). \quad (8)$$

By the classical Euler-Maclaurin identity, $\sum_{m=1}^{\infty} (\zeta(2m+1) - 1) \equiv 1/4$, confirming the exact factor-of-four homomorphism. $\square$

3 Axis II: Transverse Curvature Anchor and the UV Wall

To constrain high-frequency transverse fluctuations on the discrete lattice, we introduce a radial shell curvature metric.

Definition 3.1 (Transverse Radial Curvature Operator). On the discrete sequence space, the transverse radial curvature potential $\Delta_{\perp}^2(K)$ is defined for radial shell mode $K \geq 2$ by:

$$\Delta_{\perp}^2(K) \equiv \frac{(K-1)^2}{K^{K+1}}. \quad (9)$$

Remark 3.2 (Geometric Motivation). The form of $\Delta_{\perp}^2(K)$ arises naturally from discretized conformal geometry on radial shells: the numerator $(K-1)^2$ models quadratic boundary deformation stress between adjacent shells, while the denominator K^{K+1} reflects the super-exponential volume suppression of phase space across higher radial modes.

Theorem 3.3 (Rational Geometric Anchor). *The ground radial shell mode $K = 2$ anchors the transverse curvature to the exact rational value:*

$$\Delta_{\perp}^2(2) = \frac{(2-1)^2}{2^{2+1}} = \frac{1}{8} \equiv 0.1250000000. \quad (10)$$

Furthermore, the base mode dominates the total transverse curvature of the infinite manifold:

$$\frac{\Delta_{\perp}^2(2)}{\sum_{K=2}^{\infty} \Delta_{\perp}^2(K)} \approx 67.83\%, \quad (11)$$

while the asymptotic regime erects an impenetrable super-exponential barrier:

$$\Delta_{\perp}^2(K) \sim e^{-K \ln K} \rightarrow 0 \quad (K \rightarrow \infty). \quad (12)$$

Proof. Substituting $K = 2$ into Eq. (9) immediately yields $1/8$. For $K \geq 3$, taking logarithms gives $\ln \Delta_{\perp}^2(K) = 2 \ln(K-1) - (K+1) \ln K = -K \ln K - \ln K + 2 \ln(K-1) \sim -K \ln K$, establishing super-exponential suppression. $\square$

4 Axis III: Apéry Vacuum Trace Condensation

We construct the particle self-energy operator $\hat{W}_w$ on $\mathcal{H}_0$, governing the diagonal renormalization of the lattice.

Definition 4.1 (Diagonal Self-Energy Operator). The diagonal operator $\hat{W}_w : \mathcal{H}_0 \rightarrow \mathcal{H}_0$ is defined by:

$$[\hat{W}_w]_{kk} \equiv \frac{w(k)}{k^2(k^2 - 1)} = \frac{4}{k^3(k-1)^2(k+1)^2} \quad (k \geq 2). \quad (13)$$

Theorem 4.2 (Exact Apéry Trace Condensation). *The total operator trace $\mathcal{Z}_{\text{diag}} \equiv \text{Tr}(\hat{W}_w)$ exhibits complete algebraic anti-symmetric cancellation of all even-order Basel terms ($\zeta(2) = \pi^2/6$), condensing identically into the closed rational-Apéry form:*

$$\mathcal{Z}_{\text{diag}} = \sum_{k=2}^{\infty} \frac{4}{k^3(k-1)^2(k+1)^2} = 4\zeta(3) - \frac{19}{4} \approx 0.0582283997. \quad (14)$$

Proof. Applying complete partial fraction decomposition to the summand:

$$\frac{4}{k^3(k-1)^2(k+1)^2} = \frac{1}{(k-1)^2} + \frac{1}{(k+1)^2} - \frac{4}{k^2} - \frac{3}{k-1} + \frac{3}{k+1} + \frac{4}{k^3}. \quad (15)$$

Summing from $k = 2$ to ∞ term by term:

$$\sum_{k=2}^{\infty} \left[\frac{1}{(k-1)^2} + \frac{1}{(k+1)^2} - \frac{4}{k^2} \right] = \zeta(2) + (\zeta(2) - 1 - 1/4) - 4(\zeta(2) - 1) = -2\zeta(2) + \frac{11}{4}, \quad (16)$$

$$\sum_{k=2}^{\infty} \left[-\frac{3}{k-1} + \frac{3}{k+1} \right] = -3 \left(1 + \frac{1}{2} \right) = -\frac{9}{2}, \quad (17)$$

$$\sum_{k=2}^{\infty} \frac{4}{k^3} = 4(\zeta(3) - 1). \quad (18)$$

Combining the partial sums, the $\zeta(2)$ components cancel identically via the anti-symmetric pairing:

$$\mathcal{Z}_{\text{diag}} = \left(-2\zeta(2) + \frac{11}{4} \right) - \frac{9}{2} + 4\zeta(3) - 4 + 2\zeta(2) = 4\zeta(3) - \frac{19}{4}. \quad (19)$$

This confirms that the vacuum trace is purely rational and dependent solely on Apéry's constant $\zeta(3)$. $\square$

Theorem 4.3 (Extreme Infrared Basel Pinning). *The modal energy distribution $P(k) \equiv [\hat{W}_w]_{kk}/\mathcal{Z}_{\text{diag}}$ is pinned almost entirely to the fundamental doublet $\{2, 3\}$:*

$$P(2) = \frac{1/18}{\mathcal{Z}_{\text{diag}}} \approx 95.4110\%, \quad P(3) = \frac{1/432}{\mathcal{Z}_{\text{diag}}} \approx 3.9755\%, \quad (20)$$

such that the fundamental doublet captures 99.3865% of the total lattice trace measure.

Table 1: Tri-axial discrete invariants and modal energy distribution of the 3-point harmonic lattice.

Mode k	Kinetic Weight $w(k)$	Curvature $\Delta_{\perp}^2(k)$	Self-Energy $[\hat{W}_w]_{kk}$	Probability $P(k)$
2	$2/3 \approx 0.666667$	$1/8 = 0.125000$	$1/18 \approx 0.055556$	95.4110%
3	$1/6 \approx 0.166667$	$4/81 \approx 0.049383$	$1/432 \approx 0.002315$	3.9755%
4	$1/15 \approx 0.066667$	$9/1024 \approx 0.008789$	$1/3600 \approx 0.000278$	0.4770%
5	$1/30 \approx 0.033333$	$16/15625 \approx 0.001024$	$1/18000 \approx 0.000056$	0.0954%
6	$1/52.5 \approx 0.019048$	$25/279936 \approx 0.000089$	$1/66150 \approx 0.000015$	0.0260%
Doublet $\{2, 3\}$	$5/6 \approx 0.833333$	—	$25/432 \approx 0.057870$	99.3865%
Bulk $\sum_{k=2}^{\infty}$	1.000000	≈ 0.184288	$4\zeta(3) - 19/4 \approx 0.058228$	100.0000%

5 Transfer Operator Kernel Triviality via Carlson's Theorem

We examine the spectral kernel of the weighted transfer operator $\hat{T}_w(s)$ acting on $\mathcal{H}_0$.

Definition 5.1 (Weighted Transfer Operator). For complex $s = \sigma + it$, $\hat{T}_w(s) : \mathcal{H}_0 \rightarrow \mathcal{H}_0$ is defined by:

$$[\hat{T}_w(s)]_{m,k} \equiv \frac{\sqrt{w(k)}k^{-s}}{k^{m-1}} \quad (m, k \geq 2). \quad (21)$$

Theorem 5.2 (Hilbert-Schmidt Boundedness). For any $\sigma > 0$, $\hat{T}_w(s)$ belongs to the Hilbert-Schmidt ideal $\mathcal{S}_2(\mathcal{H}_0)$:

$$\|\hat{T}_w(s)\|_{\text{HS}}^2 = \sum_{m=2}^{\infty} \sum_{k=2}^{\infty} |[\hat{T}_w(s)]_{m,k}|^2 < \infty. \quad (22)$$

Proof. Directly evaluating the double Frobenius norm sum:

$$\|\hat{T}_w(s)\|_{\text{HS}}^2 = \sum_{m=2}^{\infty} \sum_{k=2}^{\infty} \frac{w(k)k^{-2\sigma}}{k^{2m-2}} = \left(\sum_{k=2}^{\infty} \frac{w(k)}{k^{2\sigma}} \right) \left(\sum_{m=2}^{\infty} \frac{1}{k^{2m-2}} \right). \quad (23)$$

Summing the geometric progression in m : $\sum_{m=2}^{\infty} k^{-2(m-1)} = \frac{1}{k^2-1}$. Substituting $w(k) = \frac{4}{k(k^2-1)}$:

$$\|\hat{T}_w(s)\|_{\text{HS}}^2 = \sum_{k=2}^{\infty} \frac{4}{k^{1+2\sigma}(k^2-1)^2} \leq \sum_{k=2}^{\infty} \frac{4}{k^{5+2\sigma}} < \infty, \quad (24)$$

which converges unconditionally for all $\sigma > 0$. $\square$

Theorem 5.3 (Strict Kernel Triviality). For any s in the critical strip $0 < \text{Re}(s) < 1$, the kernel of $\hat{T}_w(s)$ is strictly trivial:

$$\ker \hat{T}_w(s) = \{0\}. \quad (25)$$

Proof. Let $\phi = (\phi_2, \phi_3, \dots)^T \in \ell^2(\mathbb{N}_{\geq 2})$ satisfy $\hat{T}_w(s)\phi = 0$. By definition, for all $m \geq 2$:

$$\sum_{k=2}^{\infty} \frac{\sqrt{w(k)}k^{-s}}{k^{m-1}} \phi_k = 0. \quad (26)$$

Define the continuous analytic Dirichlet generator $F(z)$ on the half-plane $\text{Re}(z) \geq 2$:

$$F(z) \equiv \sum_{k=2}^{\infty} \frac{c_k}{k^z}, \quad c_k \equiv \sqrt{w(k)}k^{-s}\phi_k. \quad (27)$$

By the Cauchy-Schwarz inequality, since $\phi \in \ell^2$ and $w(k) \sim 4k^{-3}$:

$$\sum_{k=2}^{\infty} |c_k| \leq \left(\sum_{k=2}^{\infty} |\phi_k|^2 \right)^{1/2} \left(\sum_{k=2}^{\infty} w(k)k^{-2\sigma} \right)^{1/2} < \infty. \quad (28)$$

Hence, $F(z)$ is bounded and holomorphic on $\text{Re}(z) \geq 2$. Under the coordinate translation $w \equiv z - 2$, the domain maps to the standard right half-plane $\text{Re}(w) \geq 0$. By the Phragmén-Lindelöf principle, $F(w+2)$ is of exponential type $\tau = 0 < \pi$. Since $F(m-1) = 0$ for all integers $m \geq 2$ ($w = 0, 1, 2, \dots$), Carlson's theorem applies unconditionally, establishing $F(z) \equiv 0$. By the uniqueness of Dirichlet series, $c_k = 0$ for all $k \geq 2$. Because $\sqrt{w(k)}k^{-s} \neq 0$, it follows that $\phi_k = 0$ for all k . Thus, $\ker \hat{T}_w(s) = \{0\}$. $\square$

Proposition 5.4 (Arithmetic Frustration Gaps). *Prime modes generate independent multiplicative sub-lattices within $\mathbb{N}_{\geq 2}$. For any distinct primes p, q, r , the logarithmic phase difference $\Delta_{pqr} \equiv |\ln(pq/r)|$ satisfies Baker's theorem on linear forms in logarithms:*

$$\Delta_{pqr} > \frac{C(p, q, r)}{(\max\{\ln(pq), \ln r\})^d} > 0. \quad (29)$$

This positive frustration gap prevents accidental spectral mode-locking, enforcing level repulsion consistent with Gaussian Unitary Ensemble (GUE) statistics.

6 Boundary Intertwining and Coherent State Synthesis

To interface the discrete lattice with continuous boundary scattering, we introduce the extended Hilbert space $\mathcal{H}_{\text{ext}} = \mathbb{C}|1\rangle_{\text{bg}} \oplus \ell^2(\mathbb{N}_{\geq 2})$, where $|1\rangle_{\text{bg}}$ is the Archimedean vacuum mode ($w(1) \equiv 1$).

Definition 6.1 (Boundary Coherent State). The boundary coherent state vector $\Phi_w \in \mathcal{H}_{\text{ext}}$ weights the discrete kinetic modes by their Laplacian amplitudes:

$$\Phi_w \equiv |1\rangle_{\text{bg}} \oplus \sum_{k=2}^{\infty} \sqrt{w(k)}|k\rangle. \quad (30)$$

By the telescoping partition of unity (Theorem 2.2), Φ_w satisfies exact unit norm normalization:

$$\|\Phi_w\|_{\mathcal{H}_{\text{ext}}}^2 = 1 + \sum_{k=2}^{\infty} w(k) = 1 + 1 = 2 < \infty. \quad (31)$$

Definition 6.2 (Boundary Intertwining Operator). The intertwining mapping $\hat{V} : \mathcal{H}_{\text{ext}} \rightarrow L^2(\mathbb{R}_+^\times, \frac{dx}{x})$ maps the discrete state vector into continuous automorphic boundary space:

$$\hat{V}|1\rangle_{\text{bg}} \equiv x^{1/4}e^{-\pi x}, \quad \hat{V}|k\rangle \equiv \frac{1}{\sqrt{w(k)}}x^{1/4}e^{-\pi k^2 x} \quad (k \geq 2). \quad (32)$$

As detailed in Part II, the scaling factor $(w(k))^{-1/2}$ represents the local isometric measure density bridging the discrete measure $(\ell^2, \sum w_k)$ to the continuous boundary. This boundary intertwining operator synthesizes the automorphic wave profile $\Psi_w(x) = \sum e^{-\pi k^2 x}$, providing the direct geometric foundation for the two-track spectral audit and 105-DPS zero computation in Part II.

7 Conclusion of Part I

We have established the complete mathematical foundation of the 3-point harmonic discrete Laplacian lattice on $\ell^2(\mathbb{N}_{\geq 2})$. The framework is anchored by strict telescoping unit measure conservation ($\sum w = 1$), an exact 0.125 rational curvature anchor with super-exponential UV shielding, and Apéry trace condensation ($4\zeta(3) - 19/4$) enforcing 99.39% Basel doublet pinning. Applying Carlson's theorem under half-plane translation establishes the absolute absence of point-spectrum bound states ($\ker \hat{T}_w(s) = \{0\}$).

These discrete algebraic and measure-theoretic invariants provide the rigorous foundation for Part II, where we execute a dual-track spectral investigation, analyzing the non-vanishing phase alignment of $\hat{T}_w$ alongside an ultra-precision 105-DPS audit of the emergent boundary resonance zeros.

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